

Duration: 3 Hours**[Max Marks: 80]**

- (1) Question No 1 is Compulsory.
- (2) Attempt any three questions out of the remaining five.
- (3) All questions carry equal marks.
- (4) Assume suitable data, if required and state it clearly.

1 Attempt any **four** **[20]**

- a) Comment on the Representation Power of MLPs.
- b) Explain Gradient Descent in Deep Learning.
- c) Explain the dropout method and its advantages.
- d) What are Denoising Autoencoders?
- e) Explain Pooling operation in CNN.

2 a) What are the Three Classes of Deep Learning , explain each? **[10]**

b) Explain and analyze the architectural of AlexNet Convolution Neural Network. **[10]**

3 a) What are the different types of Gradient Descent methods, explain any three of them. **[10]**

b) Differentiate between the architecture of LSTM and GRU network. **[10]**

4 a) Explain the key components of an RNN. **[10]**

b) Consider a CNN layer with the following configuration: **[10]**

-The input to the layer has 32 channels and a spatial size of 64x64.

-The convolutional layer has 64 filters (kernels), each of size 3x3, with a stride of 1 and no padding.

-Each filter is applied to every channel of the input.

Calculate the total number of parameters (weights) in this convolutional layer.

5 a) Comment on the significance of Loss functions and explain different types of Loss functions while training a network. **[10]**

b) Explain any three types of Autoencoders. **[10]**

6 a) What is the significance of Activation Functions in Neural Networks, explain different types Activation functions used in NN. **[10]**

b) Explain Generative Adversarial Networks Architecture and its applications. **[10]**
